Code No: 117CD **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year I Semester Examinations, July/August - 2023
DATA WAREHOUSING AND DATA MINING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note : i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions

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PART – A

(25 Marks)

1.a)	Define OLAP Cube.	[2]
b)	List the main characteristics of a data warehouse.	[3]
c)	What is data mining?	[2]
d)	List out the main challenges in data mining.	[3]
e)	Write a short note on association rule learning.	[2]
f)	List the partition algorithms used for in association rule mining.	[3]
g)	How are classifiers evaluated?	[2]
h)	What are the advantages of using decision trees for classification?	[3]
i)	What is outlier detection?	[2]
j)	List the weaknesses of hierarchical clustering.	[3]

PART – B

(50 Marks)

2.a) Describe the logical (multi-dimensional) data modeling approach used in data Warehousing.
b) State and explain the merits and demerits of using a data warehouse in an organization. [5+5]

OR

3.a) Differentiate between fully additive, semi-additive, and non-additive measures in a data warehouse.
b) Explain the architecture of a data warehouse and its components. [5+5]

4.a) Discuss about the different approaches to handle missing data in data mining.
b) Explain the concept of dimensionality reduction and its significance. [5+5]

OR

5. Describe the various techniques used for data cleaning and data transformation in data mining? Provide examples. [10]

6. Explain the following terms.
a) Maximal frequent item set.
b) Association Rule Generation. [5+5]

OR

7. Explain the FP-Growth algorithm with any suitable example and also discuss the advantages and limitations. [10]

8.a) Explain the general approaches to solving a classification problem.
b) Describe the process of decision tree construction in classification. [5+5]

OR

9. Compare and contrast the performance and characteristics of decision trees, Naive-Bayes classifier, and K-Nearest Neighbor (KNN) classification. [10]

10. Explain the following.
a) Significance of clustering in various domains.
b) Hierarchical clustering. [5+5]

OR

11. Describe the Partitioning Around Medoids (PAM) algorithm and how it differs from K-means. [10]

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